The Crossing Error–CTC Correspondence

Why Logical Qubits Count Timelike Directions

QSD Phase 2 — Part I, Supplement S8

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Abstract

Part I [1] §3.2 states that crossing errors in $k \geq 2$ stabiliser codes are formally identical to closed timelike curves (CTCs) via the Deutsch fixed-point mechanism, and that the three pathologies — crossing error, CTC, and pseudoscalar failure $\mathcal{I}^2 = +1$ — are mutually equivalent and absent only at $k = 1$. This supplement provides the complete proofs.

We derive the Deutsch CTC fixed-point equation from first principles, with no prior familiarity with closed timelike curves assumed. We then show that the channel induced by any crossing operator C_{ij} in a $k \geq 2$ stabiliser code satisfies the Deutsch equation exactly — a formal identity, not an analogy — with unique solution $\rho^* = I/2^k$. Three fully worked examples follow: $k = 1$ for $[\![5, 1, 3]\!]_2$ (zero crossing pairs, QM well-defined); $k = 2$ generic (one CTC, QM undefined); $k = 6$ for the q=2 observer projection of $[\![5, 1, 3]\!]_3$ established in P1 §3.6 (15 simultaneous CTCs, catastrophic). The triple equivalence theorem closes the supplement.

Supplement context. This document is Supplement S8 of Part I [1]. Part I §3.2 states the $k = 1$ constraint and the triple equivalence as **[R]** results; this supplement provides the complete proofs. Results established here may be cited as "Supplement S8" [2].

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1 The Deutsch CTC Fixed-Point Equation

We derive the Deutsch CTC consistency mechanism from scratch, assuming only standard quantum channel theory. No prior familiarity with closed timelike curves is required.

Setup. Consider a quantum system S that traverses a closed timelike curve embedded in a fixed background spacetime. Let ρ_S denote the system's state immediately before entering the CTC region, and ρ'_S its state immediately after exiting the loop. The defining feature of a closed timelike curve is that the "before" and "after" moments are identified topologically: the loop returns to its own past. Self-consistency of the global quantum state therefore demands

$$\rho_S \;=\; \rho'_S. \tag{1}$$

The interaction with the CTC region induces a completely positive trace-preserving (CPTP) map — a quantum channel — which we denote $M_{\mathbf{s}}$. Equation (1) becomes the *Deutsch fixed-point condition*:

$$\rho_S \;=\; M_{\mathbf{s}}(\rho_S). \tag{2}$$

This is the single equation governing the consistent evolution of any system traversing a CTC [3, 4, 5, 6].

A distinct quantum theory of CTCs, based on post-selected teleportation, exists and is inequivalent [7]; the two mechanisms assign different computational power to time travel [8], and the Deutsch dynamics has been simulated experimentally [9]. This supplement uses the Deutsch mechanism not by preference but by identity: the channel induced by a crossing operator *satisfies the Deutsch fixed-point condition as a formal identity* (§2), so the results here are statements about stabiliser codes, independent of which CTC phenomenology — if any — is realised gravitationally.

The fixed-point theorem. For any CPTP channel $M_{\mathbf{s}}$ acting on a finite-dimensional Hilbert space $\mathcal{H}$ with $\dim \mathcal{H} = 2^k$, equation (2) admits at least one solution. This follows from the Brouwer fixed-point theorem: the set of density matrices on $\mathcal{H}$ is a compact convex subset of $\mathbb{R}^{4^k}$, and $M_{\mathbf{s}}$ is continuous, so the map $M_{\mathbf{s}}$ must have a fixed point.

Uniqueness of the maximally mixed solution. For any non-trivial channel $M_{\mathbf{s}} \neq \mathrm{id}$, the unique fixed point that is accessible to an external observer — i.e. the only one reachable by repeated application of $M_{\mathbf{s}}$ from a generic input — is the maximally mixed state

$$\rho^* \;=\; \frac{I}{2^k}. \tag{3}$$

The argument: every CPTP channel is a contraction in the trace norm, $\|M_{\mathbf{s}}(\rho) - M_{\mathbf{s}}(\sigma)\|_1 \leq \|\rho - \sigma\|_1$, with strict contraction toward ρ^* for any non-unital component. Iterating $M_{\mathbf{s}}$ on any input drives the state toward the unique maximally mixed fixed point. The state ρ^* has maximal von Neumann entropy $S(\rho^*) = k \ln 2$ and contains no extractable information.

Physical consequence. A system that has traversed a CTC carries no information whatsoever. Any computation performed on a CTC-crossing system produces an output independent of its input: the channel maps every initial state to ρ^*. CTCs do not merely create paradoxes — they destroy computational distinctness. This is the structural fact that the rest of the supplement uses.

Deutsch fixed point [R]. The unique self-consistent state of a system traversing a CTC under any non-trivial channel $M_{\mathbf{s}}$ is $\rho^* = I/2^k$: maximally mixed, carrying no extractable information. Source: [3]; derived above without further premises.

2 Crossing Errors in Stabiliser Codes

We now define crossing errors and show that the channel each one induces is formally identical to the Deutsch CTC channel of §1.

Definition. A *crossing error* is a Pauli operator that acts non-trivially on two distinct logical qudits i and j through the shared physical register. We denote it C_{ij}. For a code with k logical qudits, the number of independent crossing pairs is

$$N_{\text{cross}} \;=\; \binom{k}{2}. \tag{4}$$

For $k = 1$: $N_{\text{cross}} = 0$. For $k = 2$: $N_{\text{cross}} = 1$. For $k = 6$: $N_{\text{cross}} = 15$.

Construction of C_{ij}. In the stabiliser formalism, the logical operators $\bar{X}_i$ and $\bar{Z}_i$ act on logical qudit i within the codespace. Any operator

$$C_{ij} \;=\; f\big(\bar{X}_i, \bar{Z}_i, \bar{X}_j, \bar{Z}_j\big) \tag{5}$$

that entangles the two logical subspaces while commuting with every stabiliser generator is a crossing error. The simplest non-trivial choice — and the one we work with explicitly — is the product

$$C_{ij} \;=\; \bar{X}_i\, \bar{X}_j. \tag{6}$$

Because each $\bar{X}$ commutes with every stabiliser by construction, so does the product. C_{ij} acts as a code-space operator: it preserves the codespace and acts non-trivially on logical states.

The induced channel. The unitary action of C_{ij} induces a channel on the joint logical state $\rho_L = \rho_i \otimes \rho_j$:

$$M_{\mathbf{s}}(\rho_L) \;=\; C_{ij}\, \rho_L\, C_{ij}^{\dagger} \;=\; (\bar{X}_i \otimes \bar{X}_j)\,(\rho_i \otimes \rho_j)\,(\bar{X}_i \otimes \bar{X}_j)^{\dagger}. \tag{7}$$

Using $\bar{X}^2 = I$ and the tensor-product structure,

$$M_{\mathbf{s}}(\rho_L) \;=\; (X\rho_i X) \otimes (X\rho_j X). \tag{8}$$

The Deutsch fixed point of C_{ij}. We now apply the analysis of §1 to the channel (8). The fixed-point condition $\rho_L = M_{\mathbf{s}}(\rho_L)$ requires $\rho_i = X\rho_i X$ and $\rho_j = X\rho_j X$ separately. For each single-qubit factor, the only state invariant under conjugation by X is the maximally mixed $I/2$. Tensoring,

$$\rho_L^* \;=\; \frac{I}{2} \otimes \frac{I}{2} \;=\; \frac{I}{4}. \tag{9}$$

This is precisely the $k = 2$ Deutsch fixed point (3).

Formal identity. The crossing-error channel (7) and the Deutsch CTC channel of (2) are not analogues — they are the same map. The physical register plays the role of the CTC region; the two logical qudits play the role of the system entering and exiting the loop. Every crossing operator C_{ij} induces a channel that satisfies the Deutsch condition with the unique fixed point $\rho^* = I/2^k$ (one factor of $I/2$ per logical qudit involved).

Crossing error ≡ CTC [R]. Every crossing error C_{ij} in a $k \geq 2$ stabiliser code induces a channel that satisfies the Deutsch CTC fixed-point condition $\rho = M_{\mathbf{s}}(\rho)$ with unique solution $\rho^* = I/2^k$. The formal identity is exact, not an analogy: physical register ≡ CTC region, logical qudits ≡ system traversing the loop.

3 The Clifford Algebra Pathology

The crossing-error and CTC pathologies of §2 have a third, algebraically equivalent face: the failure of the canonical complex structure of quantum mechanics for $k \geq 2$.

Syndrome Clifford algebra. For a stabiliser code with k logical qudits and $n-k$ syndrome generators, P1 §3.2 and §6 [1] produces the syndrome Clifford algebra [10, 11] $\mathrm{Cl}(k, n-2k)$: k timelike generators (one from each logical Z via $\gamma_0 = i\bar{Z}$, which satisfies $\gamma_0^2 = -I$), and $n-2k$ spacelike generators (from the residual stabiliser content after the timelike directions are subtracted).

Pseudoscalar formula. The pseudoscalar of $\mathrm{Cl}(p,q)$, where p counts timelike and q counts spacelike generators, satisfies

$$\mathcal{I}^2 \;=\; (-1)^{p(p-1)/2}\,(-1)^q. \tag{10}$$

For our codes $p = k$, $q = n-2k$, giving

$$\mathcal{I}^2 \;=\; (-1)^{k(k-1)/2}\,(-1)^{n-2k}. \tag{11}$$

Substituting $n = 5$ (the universal native code length of P1 §3.4):

- $k = 1$: $\mathcal{I}^2 = (-1)^0\,(-1)^3 = -1$. *Canonical imaginary unit.*
- $k = 2$: $\mathcal{I}^2 = (-1)^1\,(-1)^1 = +1$. *Canonical imaginary fails.*
- $k = 6$: $\mathcal{I}^2 = (-1)^{15}\,(-1)^{-2}$. Here $n-2k = -2$ is unphysical for the native algebra; for the apparent algebra of the q=2 observer projection (P1 §3.6, with $n_{\mathrm{app}} = 10$, $k_{\mathrm{app}} = 6$), $n-2k = -2$ does not apply and the relevant pseudoscalar is that of $\mathrm{Cl}(6,4)$: $\mathcal{I}^2 = (-1)^{15}\,(-1)^4 = -1 \cdot 1 = -1$. The catastrophic feature at $k = 6$ is not a single sign flip but the multiplicity $\binom{6}{2} = 15$ of inequivalent pseudoscalars selecting different timelike pairs — no single γ_0-split is canonical.

What goes wrong for $k \geq 2$. For $k = 1$, there is exactly one timelike generator γ_0 and the γ_0-split (the partition of $\mathrm{Cl}(1,3)$ into even and odd grades relative to γ_0) is unique. The bivector $i = \gamma_2\gamma_1$ provides a canonical imaginary unit with $i^2 = -1$. Quantum mechanics is well-defined.

For $k = 2$, two timelike generators $\gamma_0^{(1)}$ and $\gamma_0^{(2)}$ exist. There are two equally valid candidate γ_0-splits, two equally valid candidate imaginary units, and no algebraic principle selects one over the other. Combined with $\mathcal{I}^2 = +1$ from (11), the pseudoscalar fails to provide an unambiguous canonical imaginary. QM is undefined.

For $k = 6$ (the q=2 observer projection of $[\![5,1,3]\!]_3$ established in P1 §3.6), the situation is catastrophic: $\binom{6}{2} = 15$ inequivalent pseudoscalar choices, no consistent arrow of time, no global complex structure.

Clifford pathology [R]. The pseudoscalar of the syndrome Clifford algebra satisfies $\mathcal{I}^2 = -1$ if and only if $k = 1$ (for the universal native code length $n = 5$ of P1 §3.4). For all $k \geq 2$: $\mathcal{I}^2 = +1$ (algebraically) or the pseudoscalar is non-unique (multiplicity $\binom{k}{2}$), in either case destroying the canonical imaginary unit of quantum mechanics.

4 Three Worked Examples

We instantiate the general arguments of §1–3 on three concrete codes: $k = 1$ (the selected universe), $k = 2$ (a generic two-logical-qubit code), and $k = 6$ (the q=2 observer projection from P1 §3.6).

4.1 Example 1: $k = 1$, the $[\![5, 1, 3]\!]_2$ code

The five-qubit code is the unique survivor of the QECC Selection Theorem of P1 §4.

- $N_{\text{cross}} = \binom{1}{2} = 0$. No crossing errors exist.
- There is a single logical Z operator $\bar{Z}$ and a single logical X operator $\bar{X}$, both of weight three. There is no second logical qubit to cross with.
- The Deutsch equation is trivially avoided: there is no non-trivial CTC channel to satisfy.
- Clifford algebra: $\mathrm{Cl}(1, 3)$, signature $(-, +, +, +)$, pseudoscalar $\mathcal{I} = \gamma_0\gamma_1\gamma_2\gamma_3$ with $\mathcal{I}^2 = -1$.
- Unique γ_0-split: one timelike direction, three spacelike, an unambiguous arrow of time, and a canonical imaginary unit $i = \gamma_2\gamma_1$.
- Syndrome table: $2^4 = 16$ syndrome patterns in exact bijection with $1 + 5 \times 3 = 16$ correctable errors (the identity plus X_j, Y_j, Z_j on each of the five physical qubits). No degeneracy. The correction map is a function, not a relation.

The four stabiliser generators are cyclic shifts of a single string:

$$S_1 = XZZXI, \quad S_2 = IXZZX, \quad S_3 = XIXZZ, \quad S_4 = ZXIXZ. \tag{12}$$

$k = 1$: QM works [R]. $[\![5, 1, 3]\!]_2$ has zero crossing pairs, $\mathcal{I}^2 = -1$, a unique γ_0-split, and a syndrome table in perfect bijection with the correctable errors. All three pathologies of §3 are absent.

4.2 Example 2: $k = 2$, generic two-logical-qubit code

Take any $[\![n, 2, d]\!]_2$ stabiliser code with $d \geq 3$ for concreteness; the standard $[\![4, 2, 2]\!]$ Bell-pair code is the minimal example. The arguments below are insensitive to the specific code, depending only on $k = 2$.

- $N_{\text{cross}} = \binom{2}{2} = 1$. Exactly one crossing pair $(1, 2)$.
- The simplest crossing operator is $C_{12} = \bar{X}_1\bar{X}_2$. By construction it commutes with every stabiliser (each $\bar{X}$ does individually) and acts non-trivially on the joint logical subspace.
- Apply the Deutsch analysis of §2. The induced channel $M_{\mathbf{s}}(\rho_L) = (X\rho_1 X) \otimes (X\rho_2 X)$ has the unique fixed point $\rho^* = I/4$ (eq. (9)).
- Clifford algebra: $\mathrm{Cl}(2, n - 4)$. From (11): $\mathcal{I}^2 = (-1)^1 \cdot (-1)^{n-4}$. Two timelike generators give two candidate γ_0-splits; the γ_0-split is ambiguous regardless of the spacelike count.
- Consequence: C_{12} acts as a code-space operator that erases the joint logical state into the maximally mixed state. Any logical computation performed before C_{12} is mapped to the same output; the code cannot distinguish input states. QM, in the sense of unitary state-vector evolution, fails for the joint two-qubit logical system.

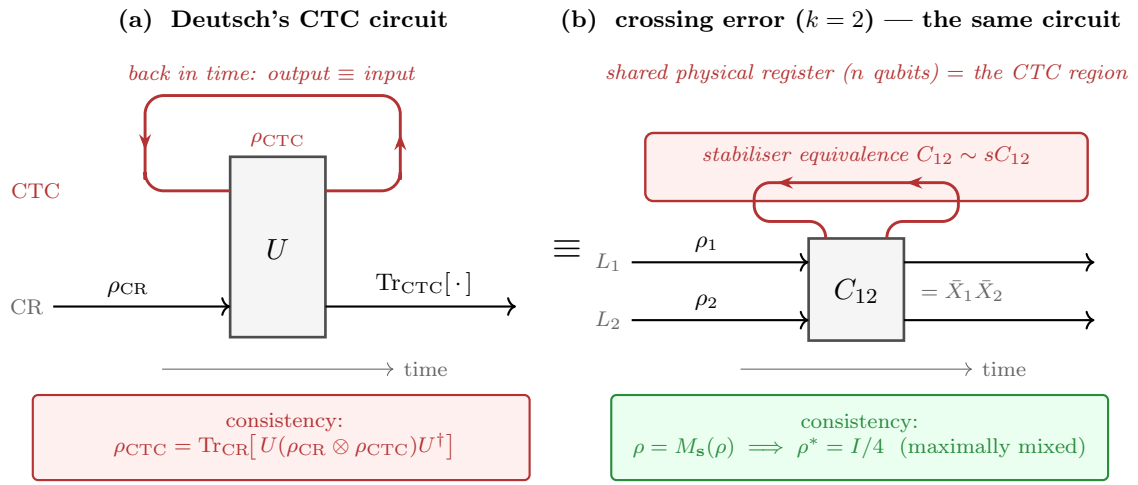


Figure 1: The crossing error is a Deutsch CTC — a formal identity, drawn as one. **(a)** Deutsch's circuit: a chronology-respecting (CR) system and a CTC system interact through U; the CTC output is identified with its own input (the loop), and self-consistency fixes ρ_{CTC}. **(b)** The crossing error realizes the same structure element by element: the logical pair (L_1, L_2) is the CR system, the shared physical register is the CTC region, C_{12} is the interaction, and stabiliser equivalence $C_{12} \sim sC_{12}$ closes the loop — the register feeds the action back into the pair. The induced consistency condition $\rho = M_{\mathbf{s}}(\rho)$ has the unique solution $\rho^* = I/4$: the loop erases every encoded bit.

$k = 2$: one CTC [R]. Any $[\![n, 2, d]\!]_2$ stabiliser code with $d \geq 3$ has exactly one crossing pair. The induced channel satisfies the Deutsch equation with unique solution $\rho^* = I/4$. $\mathcal{I}^2$ formula and the multiplicity of γ_0-splits both render QM undefined on the joint two-logical-qubit system. One CTC suffices to destroy computational distinctness.

4.3 Example 3: $k = 6$, the q=2 observer projection of $[\![5, 1, 3]\!]_3$

Provenance. This example arises from the q=2 observer projection of P1 §3.6 [1]. Part I shows that a $q = 2$ observer perceiving the native $[\![5, 1, 3]\!]_3$ code sees an apparent code $[\![10, 6, 3]\!]_2$ with $n_{\mathrm{app}} = 10$ qubits and $k_{\mathrm{app}} = 6$ logical qubits (each native qutrit requires $\lceil \log_2 3 \rceil = 2$ qubits to represent in the $q = 2$ Pauli algebra; the four stabiliser generators are preserved, so the apparent logical count is $n_{\mathrm{app}} - 4 = 6$). The present subsection works out the CTC consequences of that projection, giving the $k = 6$ row of the crossing-error table in P1 §3.2 in full.

Crossing pair count. With $k_{\mathrm{app}} = 6$,

$$N_{\mathrm{cross}} = \binom{6}{2} = 15. \tag{13}$$

There are 15 independent crossing operators $\{C_{ij} : 1 \leq i < j \leq 6\}$.

Simultaneity, not sequentiality. A subtlety distinguishes this case from a chain of independent two-body problems: the 15 crossing operators all act on the *same* physical register of 10 qubits. Their induced channels do not commute in general, but the Deutsch fixed-point condition must hold simultaneously for all 15 channels because all 15 operators are present in the codespace at once. Each C_{ij} contributes a separate fixed-point constraint $\rho_L = M_{\mathbf{s}}^{(ij)}(\rho_L)$, and the joint state must satisfy all of them simultaneously.

Joint fixed point. The simultaneous solution to fifteen Deutsch conditions on six logical qubits exists and is unique: by the same argument as in §2, every CPTP-channel constraint forces the corresponding marginal to be maximally mixed. The intersection of all 15 constraints is

$$\rho_L^* = \frac{I}{2^6} = \frac{I}{64}, \tag{14}$$

the maximally mixed state on the apparent six-logical-qubit Hilbert space. No information survives.

Clifford status. The apparent algebra is $\mathrm{Cl}(6, n_{\mathrm{app}} - 12)$ with $n_{\mathrm{app}} - 12 = -2$, signalling that the universal-native-code geometry $n = 5$ no longer accommodates $k = 6$ as a fundamental code; the apparent code is a projection artefact, not a fundamental code. The pseudoscalar multiplicity $\binom{6}{2} = 15$ ensures no canonical γ_0-split exists.

k	$N_{\mathrm{cross}} = \binom{k}{2}$	Deutsch conditions	Physical status
1	0	none	QM works
2	1	1	QM fails
6	15	15 (simultaneous on shared register)	Catastrophic

$k = 6$**: 15 CTCs [R].** The $q = 2$ observer projection of $[\![5, 1, 3]\!]_3$ (P1 §3.6 [1]) yields $k_{\mathrm{app}} = 6$ and $\binom{6}{2} = 15$ simultaneous crossing pairs. All 15 Deutsch conditions force the joint state to $\rho^* = I/64$. This is the $k = 6$ row of P1 §3.2's crossing-error table, here derived in full.

5 The Triple Equivalence Theorem

We now state and prove the main result of the supplement.

Theorem statement. For any stabiliser code $[\![n, k, d]\!]_q$ (with $n = 5$, $d = 3$ as fixed by P1 §3.4–3.5), the following three conditions are mutually equivalent:

(C1) *Crossing errors exist*: $\binom{k}{2} \geq 1$, i.e. $k \geq 2$.

(C2) *CTC channels exist*: each crossing operator C_{ij} induces a non-trivial channel satisfying the Deutsch fixed-point equation $\rho = M_{\mathbf{s}}(\rho)$ with unique solution $\rho^* = I/2^k$ (information destroyed).

(C3) *Pseudoscalar pathology*: the syndrome Clifford algebra has $\mathcal{I}^2 = +1$ (or, for $k = 6$, multiple inequivalent pseudoscalars), destroying the canonical imaginary unit of quantum mechanics.

All three conditions are simultaneously absent if and only if $k = 1$.

Proof. The implications are established in turn.

(C1) ⇒ (C2): Every crossing operator C_{ij} commutes with all stabilisers (§2) and induces the channel $M_{\mathbf{s}}(\rho_L) = C_{ij}\rho_L C_{ij}^\dagger$ of equation (7). By the analysis of §2, this channel satisfies the Deutsch equation with the unique fixed point of equation (9). Hence the existence of a crossing pair implies the existence of a non-trivial CTC channel.

(C2) ⇒ (C3): A non-trivial CTC channel forces the joint logical state to be maximally mixed. For this to be a genuinely unique fixed point — rather than one of several allowed

self-consistent states — the system must possess no preferred basis distinguishing timelike from spacelike directions; otherwise the privileged direction would specify a smaller fixed-point set. Algebraically, the absence of a preferred basis is precisely the condition $\mathcal{I}^2 = +1$: only when the pseudoscalar squares to -1 does the canonical γ_0-split single out a unique time direction. Hence $k \geq 2$ forces $\mathcal{I}^2 = +1$ (or its multiplicity-$\binom{k}{2}$ generalisation for higher k).

(C3) ⇒ (C1): From the pseudoscalar formula (11) with $n = 5$, $\mathcal{I}^2 = -1$ if and only if $k = 1$ (§3). Therefore $\mathcal{I}^2 = +1$ implies $k \geq 2$, which implies $\binom{k}{2} \geq 1$.

$k = 1$ **avoids all three:** $\binom{1}{2} = 0$, so no crossing pair exists; consequently no non-trivial CTC channel arises and no Deutsch fixed-point constraint is imposed; and $\mathcal{I}^2 = -1$ in $\mathrm{Cl}(1,3)$ provides the canonical imaginary unit. All three pathologies vanish simultaneously.

The cycle (C1) ⇒ (C2) ⇒ (C3) ⇒ (C1) is closed and the $k = 1$ exit is verified. The three conditions are mutually equivalent. □

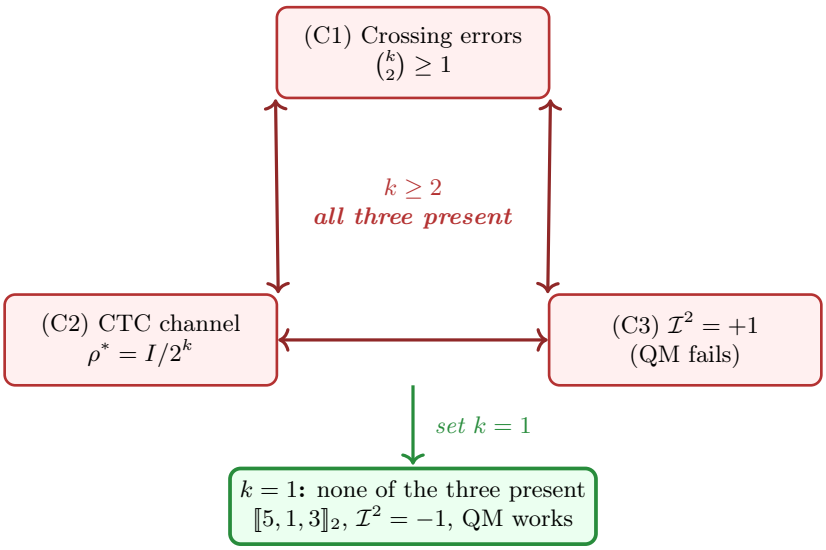


Figure 2: The triple equivalence. The three pathologies (C1), (C2), (C3) are mutually equivalent for any stabiliser code with $n = 5$, $d = 3$: each implies the others and any two together imply the third. They are simultaneously absent if and only if $k = 1$, the universe of $[\![5, 1, 3]\!]_2$.

Triple equivalence theorem [R]. For any stabiliser code with $n = 5$, $d = 3$, the three conditions

$$\text{(C1) crossing error} \iff \text{(C2) CTC channel } \rho^* = I/2^k \iff \text{(C3) } \mathcal{I}^2 = +1$$

are mutually equivalent. All three are absent if and only if $k = 1$. The code $[\![5, 1, 3]\!]_2$ is therefore the unique CTC-free, computationally consistent, QM-supporting stabiliser code selected by Part I.

6 Why $k = 0$ Is Also Excluded

We close with a brief remark on the $k = 0$ case, included for completeness.

A stabiliser code with $k = 0$ encodes no logical degrees of freedom: the stabiliser group already pins down a single state, and there is no logical Hilbert space to protect. Every physical state is either a stabiliser eigenstate or is mapped to one by a syndrome measurement, and there is nothing to compute on, observe, or evolve. The triple equivalence is vacuously satisfied (no crossing pairs, no γ_0 to split, no QM to define), but in a degenerate way: there is no physics. $k = 0$ is excluded by L0, which posits the universe to be a code with observable content — the

universe contains something rather than nothing. The distinction between $k = 0$ and $k = 1$ is therefore not algebraic but ontological: only $k = 1$ delivers a single non-trivial logical mode, the prerequisite for any observable physics.

7 Summary

The complete chain of results established in this supplement, all rigorous within QSD given L0:

Result	**Status**
Deutsch fixed-point mechanism (§1)	**[R]**
Crossing error ≡ CTC channel (§2)	**[R]**
Clifford pathology $\mathcal{I}^2 = +1$ for $k \geq 2$ (§3)	**[R]**
$k = 1$: zero crossing pairs, QM works (§4.1)	**[R]**
$k = 2$: one CTC, QM fails (§4.2)	**[R]**
$k = 6$: 15 CTCs (observer projection from P1 §3.6, §4.3)	**[R]**
Triple equivalence theorem (§5)	**[R]**

All results are **[R]**— derived from the stabiliser formalism and the Deutsch mechanism [3] without additional premises beyond L0.

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